THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two concentric circles have radii 21 cm and 35 cm (take $\pi = \frac{22}{7}$). The area of the ring between them is (cm^2): 1

(a) 2465 **(b)** 2466 **(c)** 2464 **(d)** 2463

2. A wheel makes 250 revolutions to cover 440 m (take $\pi = \frac{22}{7}$). Its radius (cm) is: 1

(a) 29 **(b)** 27 **(c)** 28 **(d)** 30

3. The area of a rhombus of side 17 cm and one diagonal 16 cm is (cm^2): 1

(a) 240 **(b)** 239 **(c)** 242 **(d)** 241

4. A triangle has sides 15 cm, 15 cm, 24 cm. The length of the shortest altitude (cm) is: 1

(a) 10 **(b)** 9 **(c)** 11 **(d)** 8

5. A rope of length 176 cm (take $\pi = \frac{22}{7}$) forms a circle. The same rope forms a square. The area of the square (cm^2) is: 1

(a) 1935 **(b)** 1936 **(c)** 1937 **(d)** 1938

6. If the radius of a circle is doubled, its circumference becomes: 1

(a) three times **(b)** four times **(c)** two times **(d)** unchanged

7. The area of a cyclic quadrilateral with sides 4 cm, 11 cm, 19 cm and 24 cm is (cm^2): 1

(a) 150 **(b)** 152 **(c)** 149 **(d)** 151

8. A circle of radius 21 cm (take $\pi = \frac{22}{7}$) is inscribed in a square. The area of the square not covered by the circle is (cm^2): 1

(a) 380 **(b)** 377 **(c)** 378 **(d)** 379

9. **Assertion (A):** The area of a parallelogram is base $\times$ corresponding height. 1
Reason (R): The area of a parallelogram is the product of two adjacent sides.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The area of a circle of radius 7 cm is 44 cm^2 (take $\pi = \frac{22}{7}$). 1
Reason (R): The area of a circle of radius r is πr^2.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A circle of radius 14 cm is inscribed in a square (take $\pi = \frac{22}{7}$). Find the area of the square left uncovered by the circle. 2

12. The area of a rhombus is 96 cm^2 and one diagonal is 16 cm. Find the other diagonal and the side of the rhombus. 2

13. Find the area of a cyclic quadrilateral with sides 4, 11, 19 and 24 cm. 2

14. A circle of radius 21 cm is inscribed in a square (take $\pi = \frac{22}{7}$). Find the area of the square left uncovered by the circle. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rope 132 cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. 3

16. A running track is made of a rectangle 70 m $\times$ 42 m with a semicircle (radius 21 m) on each of the two shorter sides (take $\pi = \frac{22}{7}$). Find the length of the track along its boundary and the area enclosed. 3

17. The sides of a triangle are 17 cm, 25 cm and 26 cm. Find its area and the altitudes on the sides 17 cm and 25 cm. 3

OR

A rope 176 cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. 3

18. A running track is made of a rectangle 90 m $\times$ 28 m with a semicircle (radius 14 m) on each of the two shorter sides (take $\pi = \frac{22}{7}$). Find the length of the track along its boundary and the area enclosed. 3

OR

The sides of a triangle are 20 cm, 21 cm and 29 cm. Find its area and the altitudes on the sides 20 cm and 21 cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A triangular plot ABC belonging to a farmer has sides AB $= 13$ m, BC $= 14$ m and CA $= 15$ m. 5

(i) Find the semi-perimeter of the plot.
(ii) Find its area using Heron's formula.
(iii) Find the altitude on the side BC.
(iv) Find the cost of levelling the plot at Rs 30 per m^2 and the altitude on the side CA.

20. In quadrilateral $ABCD$, $AB = 9$ cm, $BC = 12$ cm, $CD = 20$ cm, $DA = 25$ cm and $\angle ABC = 90^\circ$. Find the area of $ABCD$. 5

OR

A park is a rectangle 70 m $\times$ 50 m. A circular flower bed of radius 14 m (take $\pi = \frac{22}{7}$) is in it, and a path 5 m wide runs round the park inside its boundary. Find the area of the park excluding the path, the area of the grass if the bed is removed from this, and the cost of turfing the grass at ₹4 per m^2. 5

END OF QUESTION PAPER — Total: 40 Marks